

Pablo Leyva

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EDUCATION

New Jersey Institute of Technology

B.S. in Applied Mathematics & Computer Science

Newark, NJ

Aug 2023 – May 2027

- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Database Systems, Intro to Machine Learning, Neurosymbolic AI, Explainable AI, Deep Learning, Reinforcement Learning

EXPERIENCE

Algorithms and Architectures for Reasoning and Intelligent Automation Lab

Sep 2026 – Present

Professor Kieran Murphy, Professor Shivvrat Arya

Newark, NJ

- Building a semantic auditing system by comparing natural-language statements with LLM-generated first-order logic to detect semantic errors before they propagate into symbolic reasoning
- Investigating mechanistic signatures of misalignment in LLMs by combining causal circuit analysis with persistent homology over attention graphs and residual-stream activations, test detectab under distribution shift and monitor-targeted attacks

Apple

May 2026 – August 2026

AI/ML Intern

Cupertino, CA

- First intern to work on **Agentic Commerce Siri** by utilizing Apple Pay and Wallet APIs to execute multi-step agentic workflows for purchasing, booking, reserving, and scheduling and delegated sub-agents for "offline" work
- Implemented **SpeechAnalyzer** to extract user intent and used **on-device SLM** to generate user-context aware instruction prompt
- Built synthesizer for multimodal input from speech and **On-Screen Awareness** to predicting user problem and intent

President

Sep 2026 – Present

Society of Hispanic Professional Engineers (SHPE)

Newark, NJ

- Led a team of 13 officers to organize close to **100 events year** round leading to highest membership count at **250+ members**
- Invited to intern at **Angeles Investors** where I analyzed **200 startups** and led **500k** pre-seed deal for **48M** pre-money valuation
- Launched **internSHPE** mentorship program pairing underclassmen with the presidential team; all 3 mentees joined the executive board

Apple

May 2025 – August 2025

AI/ML Intern

Cupertino, CA

- Deployed a **LangGraph agent** integrating **RAG** with **MCP** to reason over technical docs for Apple Pay Merchant Partners
- Utilized qdrant for multilingual similarity search, and **GraphQL** to efficiently fetch and manipulate data across internal systems
- Built the team's **LLM evaluation pipeline** benchmarking agent trajectories, tool-use success, groundedness, hallucination rate

Caterpillar Inc.

May 2024 – August 2024

Software Engineering Intern

Chicago, IL

- Implemented data validation, transformation, and storage in **Postgres** for Autonomous Vehicle Fleet Management via Apache Airflow
- Fine-tuned **transformer models** reaching **89%** classification accuracy for predictive maintenance of heavy machinery
- Served **end-to-end data pipeline** and model predictions via **FastAPI microservices** at sub-100ms latency for 10K+ daily requests

PROJECTS

Founder - NJIT SHPE Mobile App | *Typescript* | **React-Native**

- Built and shipped the chapter's first **React-Native** app to Android and iOS store in 27 days leading a team of 3 engineers
- Scaled to 200+ users in its first month, serving member details, event registration, mentorship program and career growth in **Postgres**
- Implemented **Push Notifications**, **In-App Messaging** and Instagram-like **Feedpage** to increase engagement and retention

Evolv | *Flask, NumPy, Pandas* | **Python**

- Placed **1st of 546** at **Hackprinceton** in Predictive intelligence, building the entire backend and predictive modeling in under 12 hours
- Built a **geospatial data-center** impact simulation platform using **Flask, NumPy, Pandas**, and U.S. Census, EIA, and weather APIs
- Modeled data-center energy impact climate-adjusted cooling, grid carbon intensity, electricity pricing, probabilistic scenarios and served the analyses in **under 5 seconds** through a **Kubernetes** pipeline with parallel API orchestration, caching, and LLM-generated reports

QuStoch | *Qiskit, Flask, scikit-learn* | **Python**

- Placed **1st of 642** at **HackNYU**, learned quantum computing, stochastic brownian motion and built the simulator in under 16 hours
- Built a hybrid quantum-classical stock market simulator by encoding Brownian-motion trajectories into quantum circuits
- Developed an ML-powered financial forecasting pipeline using Python, Random Forest, and quantum-generated stochastic features, achieving an R^2 of 0.92 while improving simulation efficiency by $5\times$.

NeuroCache: KV-Cache Memory Manager | *PyTorch, ONNX, llama_cpp, WebAssembly, Transformers.js* | **C++**

- Submitted PR to **llama_cpp**
- Implemented a **KV-cache memory manager** with attention-scored **eviction policy**, slot-based allocation, and cache-replacement algorithms for dynamic context compression; benchmarking memory efficiency and response quality across long conversation sessions

TECHNICAL SKILLS

Languages : Python, C++, JavaScript/TypeScript, Swift

Systems & Infrastructure : Docker, Git, Kubernetes, PostgreSQL, REST FastAPI microservices, Apache Airflow pipelines

ML Frameworks : PyTorch, scikit-learn, Hugging Face Transformers